Chapter 14 : Breathing and Exchange of Gases

1. Match List-I with List-II. **[NEET-2025]**

List-I	List-II
A. Emphysema	I. Rapid spasms in muscle due to low Ca++ in body fluid
B. Angina Pectoris	II. Damaged alveolar walls and decreased respiratory surface
C. Glomerulonephritis	III. Acute chest pain when not enough oxygen is reaching to heart muscle
D. Tetany	IV. Inflammation of glomeruli of kidney

Choose the correct answer from the options given below :

(1) A-III, B-I, C-IV, D-II

(2) A-III, B-I, C-II, D-IV

(3) A-II, B-IV, C-III, D-I

(4) A-II, B-III, C-IV, D-I

2. Match List-I with List-II: **[Re-NEET-2024]**

List-I	List-II
A. Residual Volume	I. Maximum volume of air that can be breathed in after forced expiration
B. Vital Capacity	II. Volume of air inspired or expired during normal respiration
C. Expiratory Capacity	III. Volume of air remaining in lungs after forcible expiration
D. Tidal Volume	IV. Total volume of air expired after normal inspiration

Choose the correct answer from the options given below:

(1) A-IV, B-III, C-II, D-I

(2) A-II, B-IV, C-I, D-III

(3) A-III, B-I, C-IV, D-II

(4) A-I, B-II, C-III, D-IV

3. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R. **[Re-NEET-2024]**

Assertion A: During the transportation of gases, about 20-25 percent of CO_2 is carried by Haemoglobin as carbamino-haemoglobin.

Reason R: This binding is related to high pCO_2 and low pO_2 in tissues.

In the light of the above statements, choose the correct answer from the options given below.

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A.

4. Match List I with List II : **[NEET-2024]**

List I	List II
A. Expiratory capacity	I. Expiratory reserve volume + Tidal volume + Inspiratory reserve volume
B. Functional residual capacity	II. Tidal volume + Expiratory reserve volume
C. Vital capacity	III. Tidal volume + Inspiratory reserve volume
D. Inspiratory capacity	IV. Expiratory reserve volume + Residual volume

Choose the correct answer from the options given below :

(1) A-II, B-IV, C-I, D-III (2) A-III, B-II, C-IV, D-I

(3) A-II, B-I, C-IV, D-III (4) A-I, B-III, C-II, D-IV

5. Which of the following factors are favourable for the formation of oxyhaemoglobin in alveoli?

(1) High pO_2 and High pCO_2 **[NEET-2024]**

(2) High pO_2 and Lesser H^+ concentration

(3) Low pCO_2 and High H^+ concentration

(4) Low pCO_2 and High temperature

6. Select the sequence of steps in Respiration. **[NEET-2023-MANIPUR]**

(A) Diffusion of gases (O_2 and CO_2) across alveolar membrane.

(B) Diffusion of O_2 and CO_2 between blood and tissues.

(C) Transport of gases by the blood.

(D) Pulmonary ventilation by which atmospheric air is drawn in and CO_2 rich alveolar air is released out.

(E) Utilisation of O_2 by the cells for catabolic reactions and resultant release of CO_2.

Choose the correct answer from the options given below :

(1) (B), (C), (E), (D), (A)
(2) (A), (C), (B), (E), (D)
(3) (D), (A), (C), (B), (E)
(4) (C), (B), (A), (E), (D)

7. Vital capacity of lung is _________. **[NEET-2023]**

(1) IRV + ERV
(2) IRV + ERV + TV + RV
(3) IRV + ERV + TV - RV
(4) IRV + ERV + TV

8. Which of the following is not the function of conducting part of respiratory system ? **[NEET-2022]**

(1) Inhaled air is humidified

(2) Temperature of inhaled air is brought to body temperature

(3) Provides surface for diffusion of O_2 and CO_2

(4) It clears inhaled air from foreign particles

9. Under normal physiological conditions in human being every 100 ml of oxygenated blood can deliver ___________ml of O_2 to the tissues. **[NEET-2022]**

(1) 5ml
(2) 4 ml
(3) 10 ml
(4) 2 ml

10. Which of the following statements are correct with respect to vital capacity? **[NEET-2022]**

(a) It includes ERV, TV and IRV

(b) Total volume of air a person can inspire after a normal expiration

(c) The maximum volume of air a person can breathe in after forced expiration

(d) It includes ERV, RV and IRV.

(e) The maximum volume of air a person can breath out after a forced inspiration.

Choose the correct answer from the options given below :

(1) (b), (d) and (e)
(2) (a), (c) and (d)
(3) (a), (c) and (e)
(4) (a) and (e)

11. The partial pressures (in mm Hg) of oxygen (O_2) and carbon dioxide (CO_2) at alveoli (the site of diffusion) are : **[NEET-2021]**

(1) pO_2 = 104 and pCO_2 = 40
(2) pO_2 = 40 and pCO_2 = 45
(3) pO_2 = 95 and pCO_2 = 40
(4) pO_2 = 159 and pCO_2 = 0.3

12. Select the favourable conditions required for the formation of oxyhaemoglobin at the alveoli.

(1) High pO2, low pCO2, less H+, lower temperature **[NEET-2021]**

(2) Low pO2, high pCO2, more H+, higher temperature

(3) High pO2, high pCO2, less H+, higher temperature

(4) Low pO2, low pCO2, more H+ , higher temperature

13. Identify the wrong statement with reference to transport of oxygen. **[NEET-2020]**

(1) Low pCO2 in alveoli favours the formation of oxyhaemoglobin.

(2) Binding of oxygen with haemoglobin is mainly related to partial pressure of O2.

(3) Partial pressure of CO2 can interfere with O2 binding with haemoglobin.

(4) Higher H+ conc. in alveoli favours the formation of oxyhaemoglobin.

14. Select the correct events that occur during inspiration. **[NEET-2020]**

(a) Contraction of diaphragm

(b) Contraction of external inter-costal muscles

(c) Pulmonary volume decreases

(d) Intra pulmonary pressure increases

Choose the correct answer from the options given below :

(1) only (d)

(2) (a) and (b)

(3) (c) and (d)

(4) (a), (b) and (d)

15. The Total Lung Capacity (TLC) is the total volume of air accommodated in the lungs at the end of a forced inspiration. This includes : **[NEET-2020 COVID-19]**

(1) RV; IC (Inspiratory Capacity); EC (Expiratory Capacity); and ERV

(2) RV; ERV; IC and EC

(3) RV; ERV; VC (Vital Capacity) and FRC (Functional Residual Capacity)

(4) RV (Residual Volume); ERV (Expiratory Reserve Volume); TV (Tidal Volume); and IRV (Inspiratory Reserve Volume)

16. Match the following columns and select the correct option : **[NEET-2020 COVID-19]**

Column-I	Column-II
(a) Pneumotaxic Centre	(i) Alveoli
(b) O2 Dissociation	(ii) Pons region of brain
(c) Carbonic Anhydrase	(iii) Haemoglobin
(d) Primary site of exchange of gases	(iv) R.B.C.

(1) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

(2) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(3) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

(4) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)

17. Tidal Volume and Expiratory Reserve Volume of an athlete is 500 mL and 1000 mL respectively. What will be his Expiratory Capacity if the Residual Volume is 1200 mL? **[NEET-2019]**

(1) 1500 mL (2) 1700 mL

(3) 2200 mL (4) 2700 mL

18. Select the correct statement. **[NEET-2019-ODISHA]**

(1) Expiration occurs due to external intercostal muscles

(2) Intrapulmonary pressure is lower than the atmospheric pressure during inspiration.

(3) Inspiration occurs when atmospheric pressure is less than intrapulmonary pressure.

(4) Expiration is initiated due to contraction of diaphragm.

19. The maximum volume of air a person can breathe in after a forced expiration is known as:

(1) Expiratory Capacity **[NEET-2019-ODISHA]**

(2) Vital Capacity

(3) Inspiratory Capacity

(4) Total lung Capacity

20. Which of the following options correctly represents the lung conditions in asthma and emphysema, respectively ? **[NEET-2018]**

(1) Inflammation of bronchioles; Decreased respiratory surface

(2) Increased number of bronchioles; Increased respiratory surface

(3) Increased respiratory surface; Inflammation of bronchioles

(4) Decreased respiratory surface; Inflammation of bronchioles

21. Match the items given Column I with those in Column II and select the correct option given below

Column-I	Column-II
a. Tidal volume	i. 2500-3000 mL
b. Inspiratory Reserve	ii. 1100-1200 mL volume
c. Expiratory Reserve	iii. 500-550 mL volume
d. Residual volume	iv. 1000-1100 mL

[NEET-2018]

	a	b	c	d
(1)	iii	ii	i	iv
(2)	iii	i	iv	ii
(3)	iii	iv	ii	iii
(4)	iv	iii	ii	i

22. Which of the following is an occupational respiratory disorder ? **[NEET-2018]**

(1) Anthracis

(2) Silicosis

(3) Botulism

(4) Emphysema

23. Lungs are made up of air-filled sacs, the alveoli. They do not collapse even after forceful expiration, because of: **[NEET-2017]**

(1) Inspiratory Reserve Volume
(2) Tidal Volume
(3) Expiratory Reserve Volume
(4) Residual Volume

24. Asthma may be attributed to : **[NEET- I 2016]**

(1) bacterial infection of the lungs

(2) allergic reaction of the mast cells in the lungs

(3) inflammation of the trachea

(4) accumulation of fluid in the lungs

25. The partial pressure of oxygen in the alveoli of the lungs is :- **[NEET- II 2016]**

(1) Less than that in the blood

(2) Less than that of carbon dioxide

(3) Equal to that in the blood

(4) More than that in the blood

26. Lungs do not collapse between breaths and some air always remains in the lungs which can never be expelled because :- **[NEET- II 2016]**

(1) There is a positive intrapleural pressure

(2) Pressure in the lungs is higher than the atmospheric pressure.

(3) There is a negative pressure in the lungs.

(4) There is a negative intrapleural pressure pulling at the lung walls

ANSWER KEY

1. (4)
2. (3)
3. (3)
4. (1)
5. (2)
6. (3)
7. (4)
8. (3)
9. (1)
10. (3)
11. (1)
12. (1)
13. (4)
14. (2)
15. (4)
16. (2)
17. (1)
18. (2)
19. (2)
20. (1)
21. (2)
22. (2)
23. (4)
24. (2)
25. (4)
26. (4)